

Lab 1 - Exploring the Dataset & Tableau Basics

You will start this lab by exploring the wine dataset and its attributes. As you review the data, consider what types of basic visualizations you can create in Tableau and what insights each chart might reveal.

The main goal of this lab is to help you get comfortable with creating and interpreting basic Tableau plots—such as bar charts, histograms, maps, scatter plots, and pie charts. Focus on building individual visualizations to better understand the data and lay the foundation for more advanced analysis in future labs.

You can use either Tableau Desktop or Tableau Public Online. For the online version, visit [Tableau Public](#), then select "Create > Web Authoring" to start building visualizations in your browser.

Dataset Overview

The wine dataset contains nearly 130,000 reviews from around the world, with information such as country, region, variety, description, points (ratings), and price. This enables analysis of wine characteristics, popularity, and trends across different countries and categories.

- **Source:** [Kaggle Wine Reviews](#)

We have organized the data into several files for different analysis tasks:

- `winemag-data-with-year.csv` : Includes a year extracted from the wine title for time-based analysis.
- `wine_category.csv` : Maps wine varieties into five categories: red, white, rose, sparkling, and other.

Please download the datasets from Moodle and save them somewhere easy to find. You will use these files in this and future labs.

Observe the Dataset

Before you start visualizing, take a few minutes to review the dataset and understand what each attribute means. This will help you decide what questions you can ask and what charts might be useful.

Attributes in `winemag-data-with-year.csv` :

- `country` : Country of origin
- `province` : State or province where the wine is produced
- `region_1` , `region_2` : Specific wine regions within the province
- `winery` : Name of the winery
- `designation` : Special name/title of the wine
- `variety` : Type of grape or wine
- `title` : Full wine review title (often includes vintage/year)
- `points` : Wine rating (1-100)
- `price` : Price per
- `description` : Review text
- `taster_name` , `taster_twitter_handle` : Reviewer information

Attributes in `wine_category.csv` :

- `variety` : Type of grape or wine
- `category` : Wine category (red, white, rose, sparkling, other)

Possible Visualizations:

Analysis Goal	Chart Type	Example Questions
Summary Statistics	Bar Chart	Which countries have the highest-rated wines?
Distribution	Histogram, Box Plot	What's the price distribution for red wines?
Relationships	Scatter Plot	Are higher-rated wines more expensive?
Composition	Pie Chart, Stacked Bar Chart	What's the breakdown of wine categories globally?
Hierarchy & Quantity	Treemap	Which varieties dominate each category?
Geographic Analysis	Filled Map	Where are top-rated wines produced?
Temporal Trends	Line/Area Chart	How have ratings changed over time?
Rankings	Top N Lists	What are the top 10 most-reviewed varieties?

And more ...

Note: A single question can often be visualized in multiple ways (e.g. bar vs treemap).

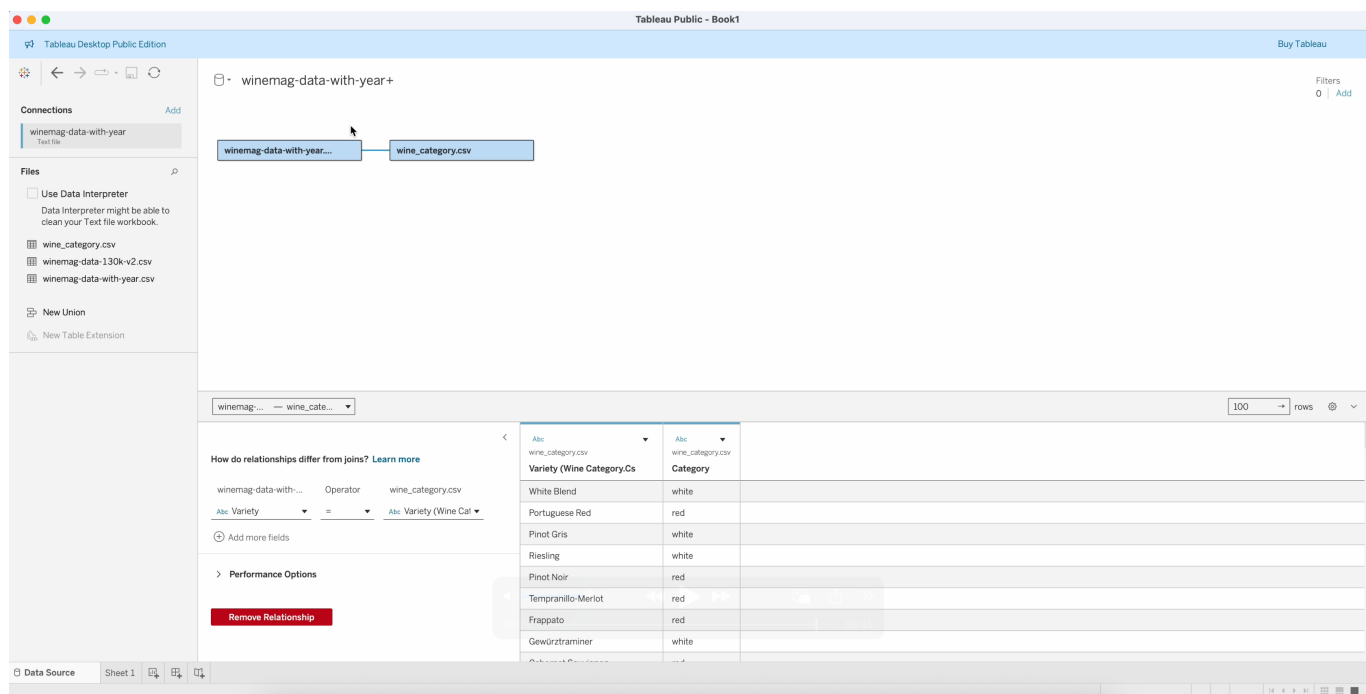
Practice in Tableau

In this section, you'll create different types of visualizations to explore the wine dataset and get comfortable using Tableau.

Goal:

- Practice importing and connecting datasets
- Create various charts (map, bar, scatter, box, line, etc.)
- Learn to format and enhance visualizations with labels, tooltips, colors, and calculated fields

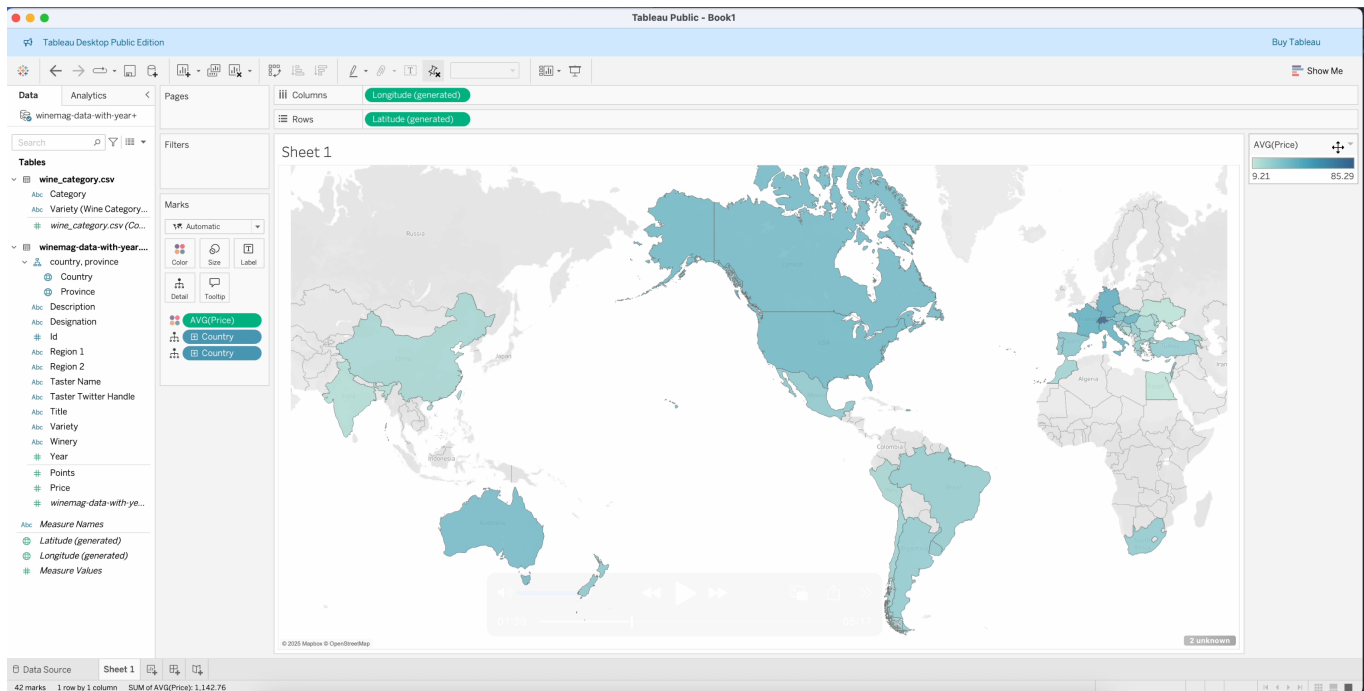
1. Import the Dataset



- Open Tableau and connect to `winemag-data-with-year.csv` as a **Text File**
- Add a second data source: `wine_category.csv`
- Use the `variety` column to **join** the two datasets

This allows us to access the wine category (red/white/sparkling/...) for each wine entry.

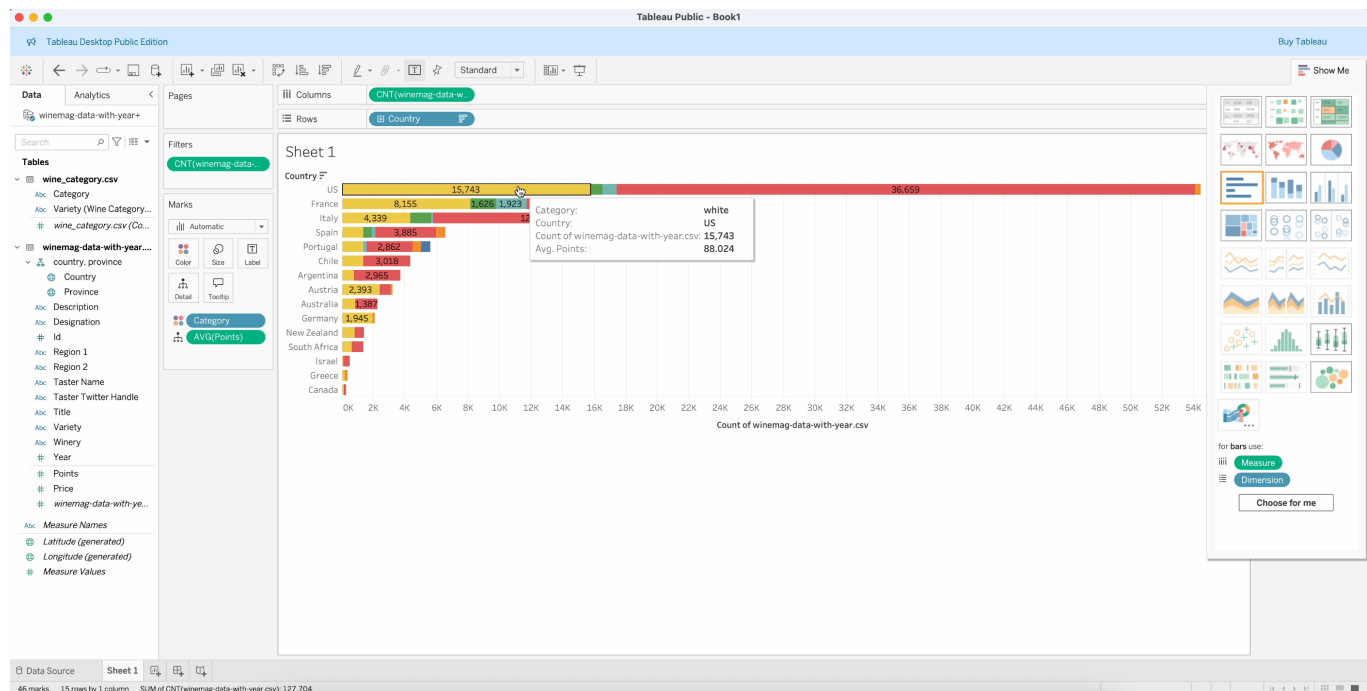
2. Map View: Wine Distribution by Country



- Double-click `country` → Tableau will auto-generate a map
- Drag attributes to customize:
 - `number of reviews (winemag-data-with-year.csv(Count))` to Color
 - (Optional) Play with other marks like Color or Size (for a symbol map). For example, add `points` (hint: to make the color representation more meaningful, click `SUM(Points)` and set Measure to Average).

You can change color palettes (click color → edit colors) and add **filters** (e.g., right-click `country` → Filter → filter by general manual selection or conditions like top 10 review counts)

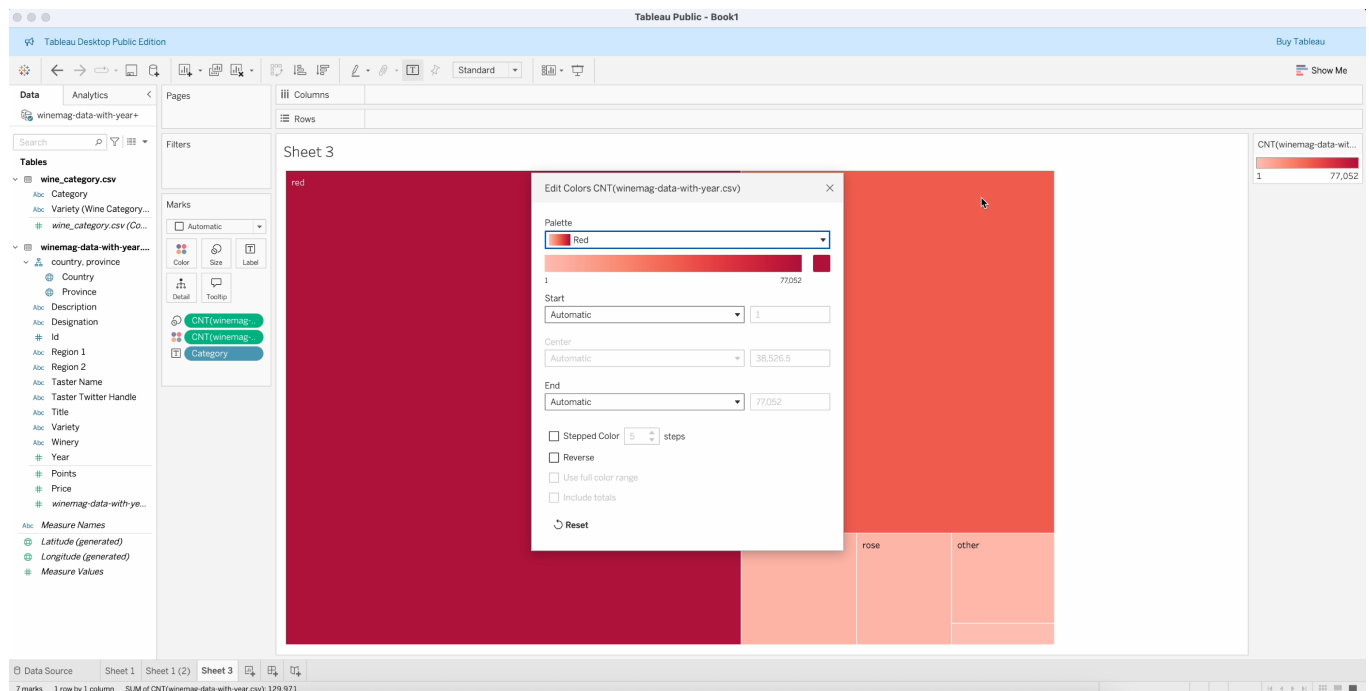
3. Bar Chart (Alternative to Map)



- In addition to a filled map, the same visualization can be shown in a bar chart
- Duplicate the map sheet and switch to a **horizontal bar chart**
- Add `category` as Color to compare wine types within each country
- Sort bars by count, and optionally drag `number of reviews` to Label to show numbers on bars

Same data, different view—more compact and easier to compare

4. Wine Category Distribution

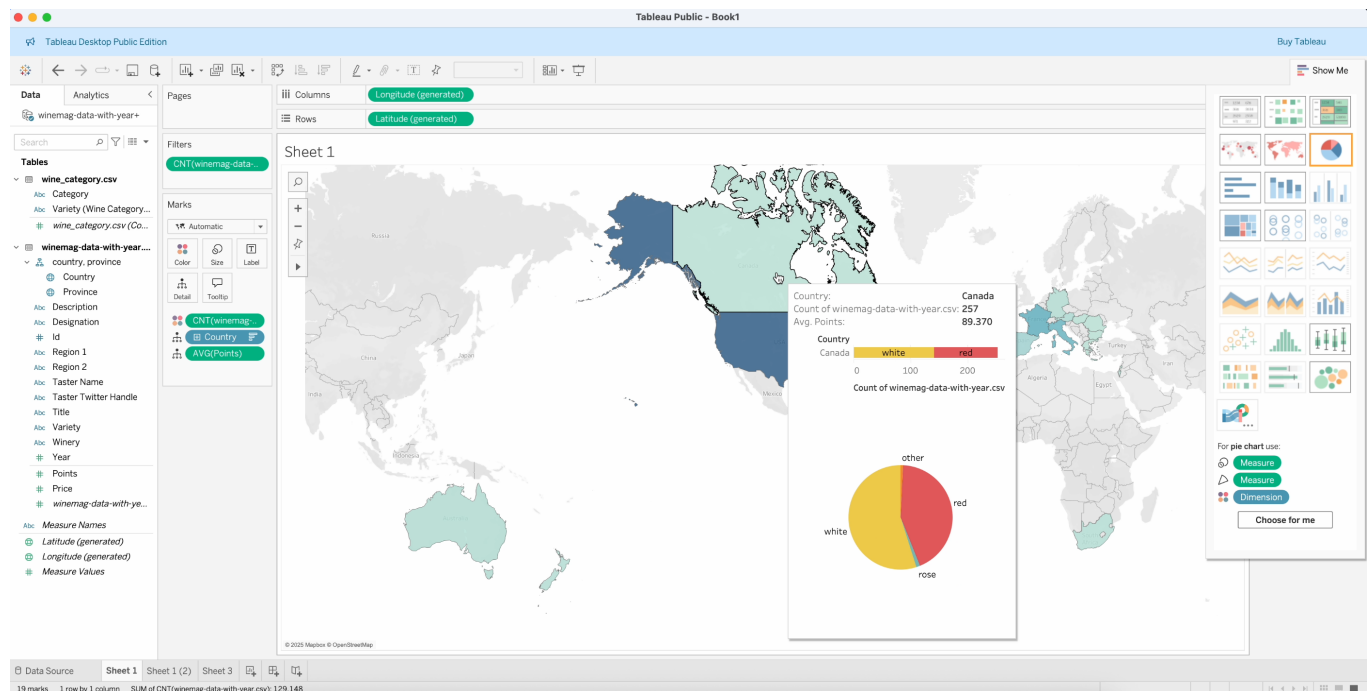


Explore how wine types are distributed globally using:

- **Treemap:** Good for comparing proportions and quantity
- **Pie chart:** Compact overview (avoid overuse)
- **Stacked bar chart:** Compare multiple categories per country or region

You can exclude unimportant or ambiguous categories like "null".

5. Add Tooltip Chart to the Map

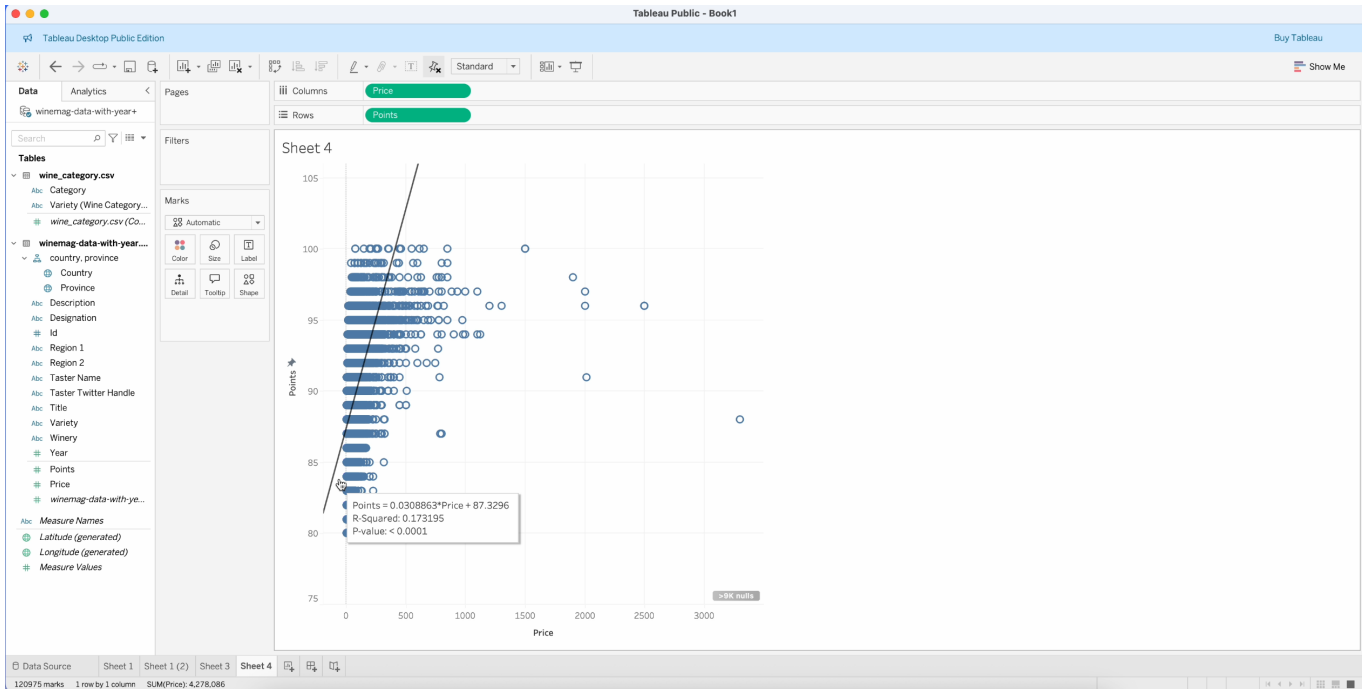


To enrich the map:

- Add the **category distribution chart** as a **Tooltip** visualization (click Tooltip → Insert → Sheets)
- When hovering over a country, the tooltip shows its wine breakdown (e.g. red vs white)

Tableau will automatically filter by the hovered country

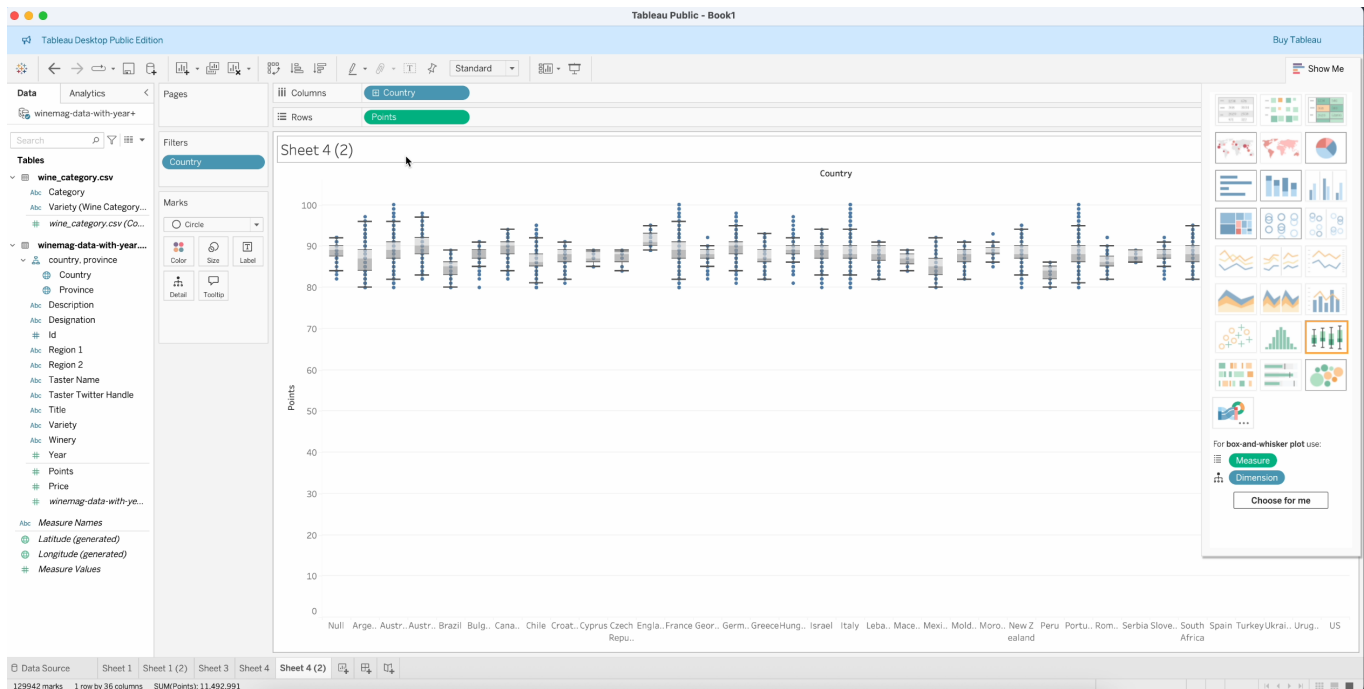
6. Statistical Relationship: Price vs. Points



- Drag `price` to Columns (X-axis) and `points` to Rows (Y-axis) to create a **scatter plot**
- In the **Analysis** pane, **uncheck "Aggregate Measures"** to display individual data points rather than summed values.
- Add `variety` as Color (optional)
- Right-click → **Show Trend Lines**

Use this to explore: Are more expensive wines rated higher?

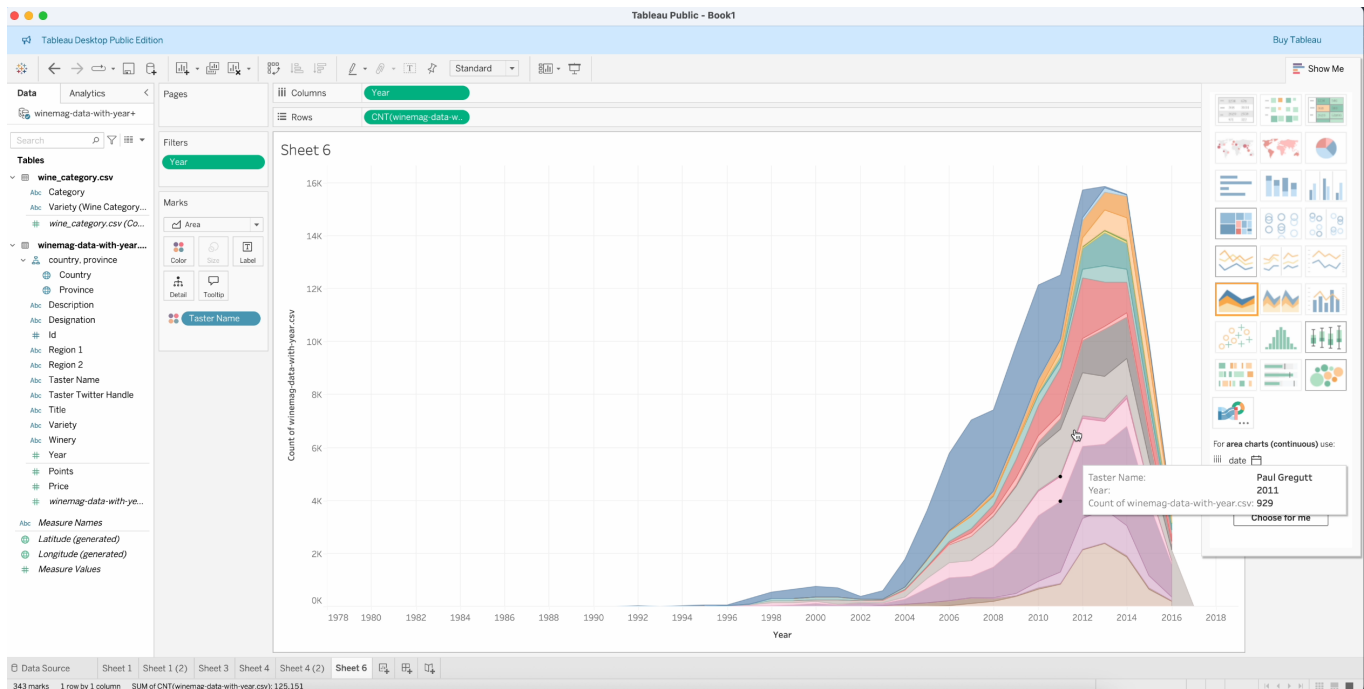
7. Box Plot: Country Comparison



- Use `points` as the measure
- Break down by `country` or `category`
- Display outliers, median, and the range of ratings. Adjust the axis range for clearer visualization and easier comparison.

Box plots offer a clear summary of variation and consistency

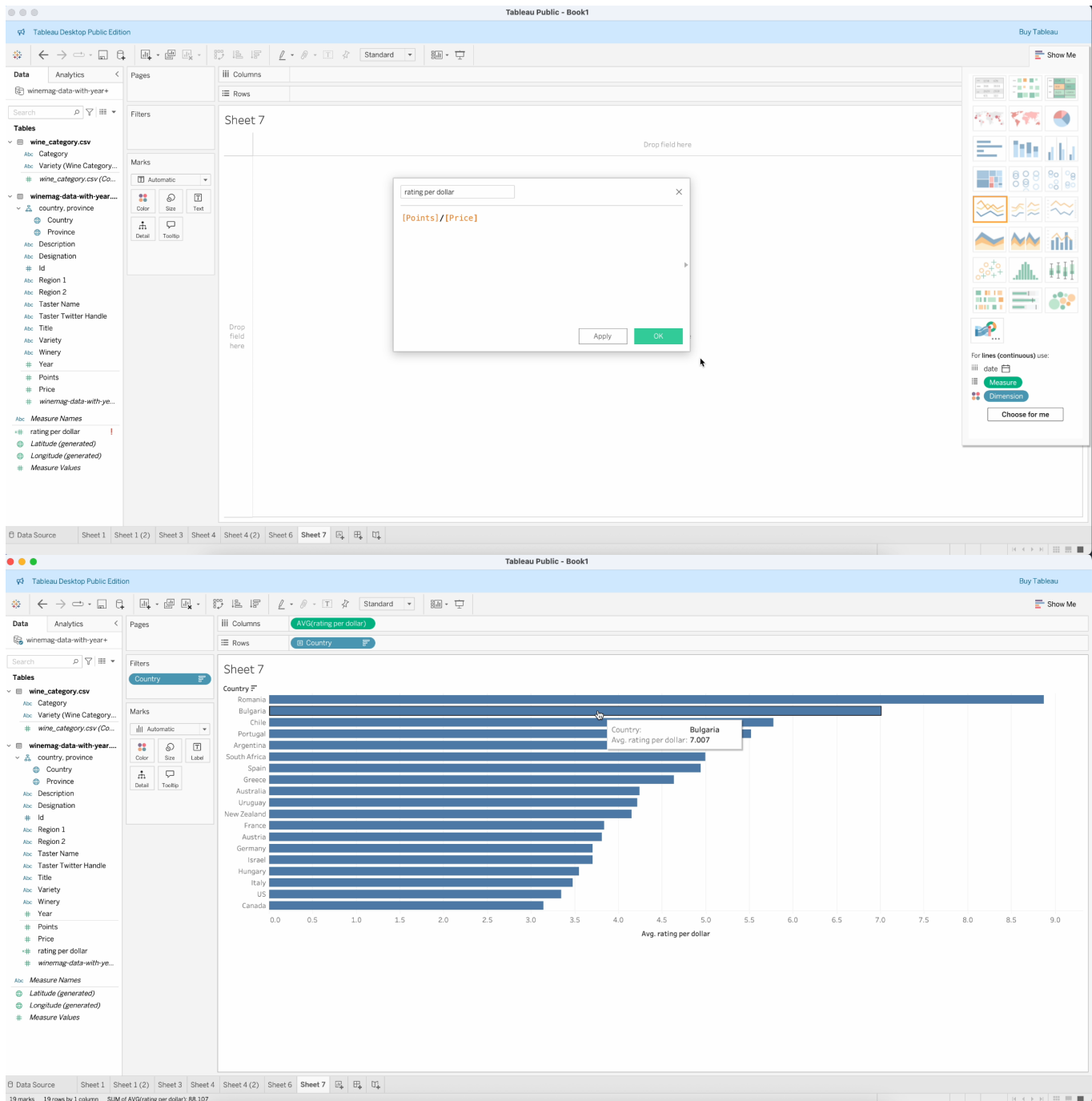
8. Time Series



- Use `year` as the Columns (X-axis). You may use filter to only keep data from 1985 to 2016 for clearer visualization.
- Show **average rating** or **review count** over time
- Try both:
 - **Line chart** with `taster_name` as color
 - **Area chart** to show contribution over time

Helps analyze trends in reviewer activity

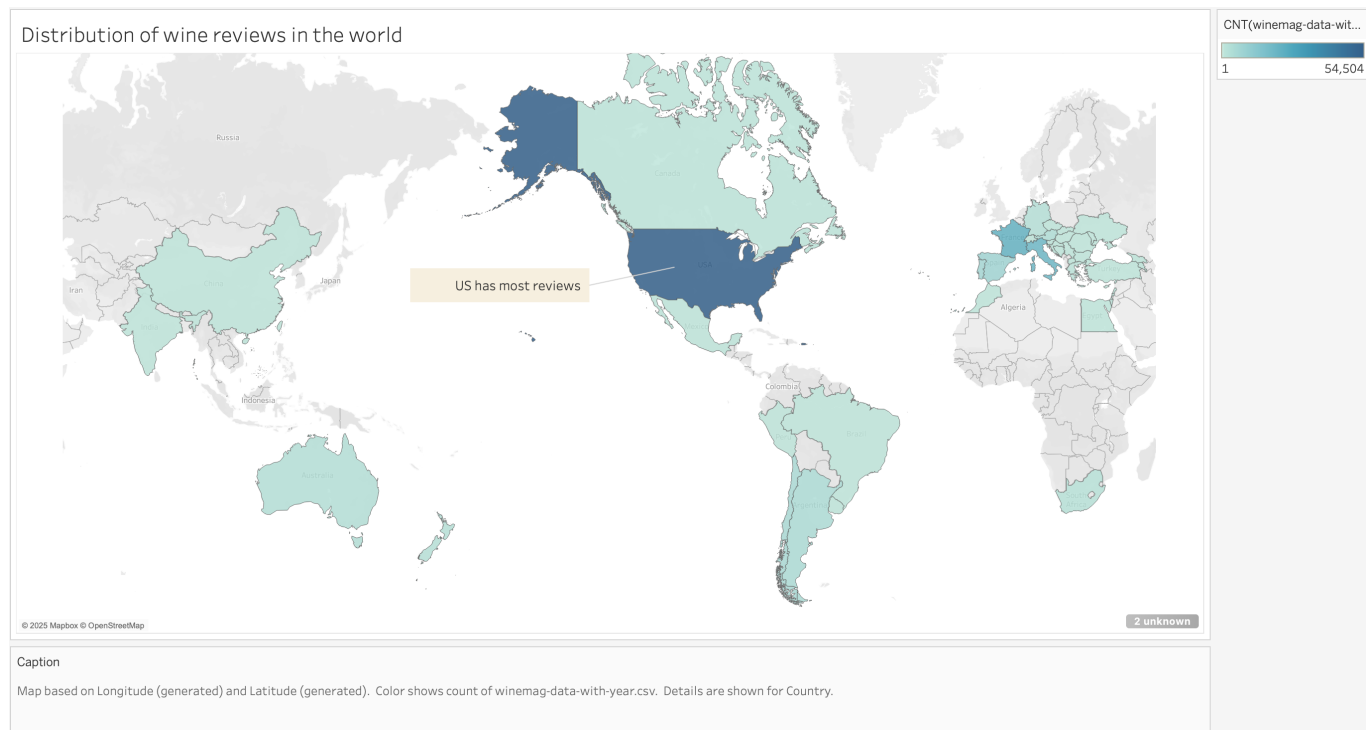
9. Calculated Field: Rating per Dollar



- On the Data panel, right-click to create a calculated field called "Rating per Dollar" with the formula $[Points]/[Price]$.
- Use this to find **high-value wines**—well-rated but affordable

A Tableau calculated field is a custom field you create by writing a formula using existing fields in your data. This lets you perform calculations, transform data (continuous to categorical), or create new metrics directly within Tableau, without changing your original dataset.

10. Final Touches: Rename, Annotate, and Caption



Make your charts easier to understand:

- **Rename** each worksheet (e.g., "Map: Avg Rating by Country"). You may rename tab names as well, which helps you distinguish different sheets when creating dashboards/stories.
- Add **annotations** to highlight interesting insights directly on your charts (right-click → Annotate). Focus on key trends, outliers, or notable comparisons.
- Enable **captions** (Worksheet → Show Caption) and customize the caption to clearly explain what each color represents, especially for charts with complex color schemes (e.g., multiple categories, gradients, or conditional formatting) when sharing.

11. Save and Publish

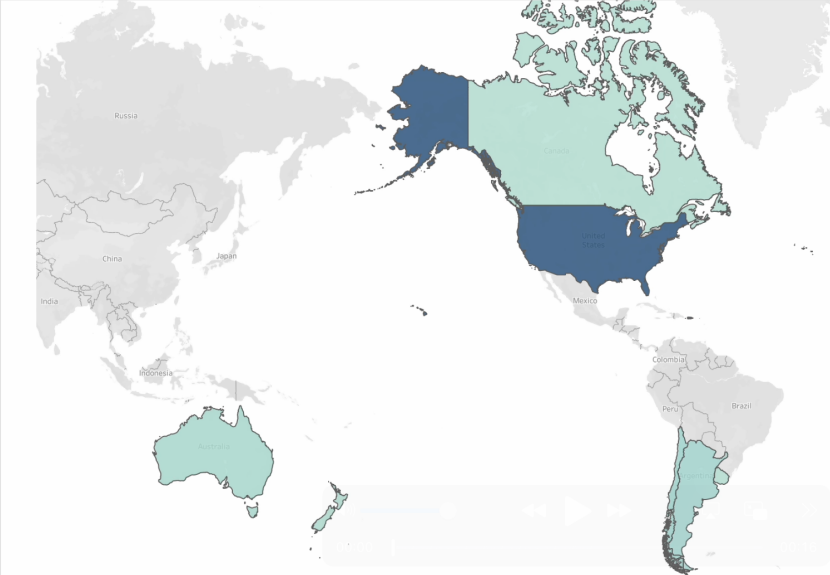
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Your summer to-do list: dive into Tableau Next, reimagine analytics, win \$45k → Join the hackathon

Wine overview by Yuran Sun

Sheet 1 Sheet 1 (2) Sheet 3 Sheet 4 Sheet 4 (2) Sheet 6 Sheet 7 Sheet 8

Distribution of reviews in world



ata-w... 54,504

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- Save your workbook to **Tableau Public**
- In Tableau Public, enable **"Show Sheets as Tabs"**. This lets others easily switch between your charts

You'll need to submit the link for your lab assignment

Your Task

Your task is to explore US wine data and practice the visualizations covered in class in Tableau. You'll build 3 individual charts that answer key questions about **wine distribution**, **wine types**, and **wine quality trends**.

Important: For all your visualizations in this lab task, you must filter the dataset to include **only wines from the United States**. In each of your chart, set a filter on the `country` field and **ONLY** keep **US**.

Please complete the following:

1. Which US State Has the Most Wine Reviews?

- Use a **map** to show wine review distribution across US states (`province`)
- Use colors to differential states by review count
- Highlight the state with the most reviews using **annotation**.

2. What Types of Wine Are Popular in Each State?

- Show wine category distribution (`category`) within each US state
- You may:
 - Create a **bar chart** for each state, using distinct colors to represent different wine categories.
 - OR present a **highlighted table** showing wine categories versus states. Use colors to indicate the popularity (i.e., count of reviews).

3. How Have US Wine Ratings Changed Over Time?

- Use a **line chart** to plot average rating (`points`) by year, using data from 1985 to 2016.
- Assign different colors for different states
- Note: you can set the y-axis (points) range from 80 to 95 for better visibility.

4. Naming and Organization:

- Rename each worksheet clearly
- Add meaningful **captions** or **annotations** for Task1 map.
- Use clean and readable formatting (labels, colors, sorting)

5. Submission:

- Save to Tableau Public
- Rename workbook: **"Wine Overview Exploration"**.
- Enable **show sheets as tabs** so all charts are visible
- **Submit your Tableau Public link on Moodle**